

University of Dhaka
 Affiliated Engineering Colleges
 Department of Computer Science and Engineering
 1st Year 1st Semester B.Sc. in CSE Final Examination, 2022
 CHE-1104: Chemistry

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) What are quantum numbers? State their significance. 4
 b) In chlorine Cl_2 gas, there is no permanent dipole-dipole interaction. Explain their absence in chlorine. 4
 c) Discuss the position of hydrogen in the periodic table. 4
 d) Explain why the reactivity of the elements increases going down Group 1 from lithium to rubidium but decreases going down Group 7 from fluorine to iodine. 2
2. a) State and Explain the mechanism of buffer solution? Give an example of a buffer solution with a pH less than 7. 6
 b) A buffer solution made by mixing together solutions containing ethanoic acid and sodium ethanoate. If you add a small amount of an acid such as dilute hydrochloric acid to this, the pH doesn't change much. Explain what happens to the extra hydrogen ions you have added? 5
 c) Give the definitions of an acid and a base on the Bronsted-Lowry theory. 3
3. a) Explain how equilibrium constant changes with temperature for exothermic and endothermic Reaction. 4
 b) State and explain the law of mass action. Derive a relationship between K_p and K_c for the reaction: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$. 4
 c) Consider the following reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, If the equilibrium constant, K_c , for this reaction is 6.0×10^2 at a certain temperature, what is the value of K_c for the reverse reaction? 4
 d) State the relationship between the faraday constant, F , the charge of the electron, e , and the Avogadro number, L . 2
4. a) With a neat and clean diagram state phase rule and explain its applications to ice-water-vapor System. What is meant by the triple point? 6
 b) During any chemical reaction, old bonds are broken and new ones made. Explain how this breaking and making of bonds leads to the change being exothermic 4
 c) Electrolytes have abnormally high value of colligative properties –Explain by giving reason. 4
5. a) Discuss the position of H in the periodic table. 5
 b) Explain “ Na^+ is possible but Na^{++} is not possible” 3
 c) Why the first ionization potential of N atom is greater than O atom. 3
 d) Why the electron affinity of Cl is more than F. 3
6. a) Deduce the rate equation for the type of following reaction: $\text{A} \rightarrow \text{products}$. 4
 b) What is meant by half- life period? Show that the half-life of second order reaction is inversely proportional to its initial concentration. 4
 c) Explain “activated complex state” with a suitable graph. 3
 d) 50% complete of a first order is need 23 minutes. what time will be need 90% completed of the reaction. 3
7. a) Explain the covalent character of ionic bond. 3
 b) Most elements have some isotopes which are radioactive. What is meant by the terms: (i) Isotopes & (ii) Radioactive 3
 c) Osmotic pressure and freezing point depression have the same origin. Explain. 4
 d) Describe a method for determining the molar mass of a non-volatile solute by the Relative lowering in vapor pressure of the solvent. 4